Reviewer Pack — Minimal Rank of Algebraic Cycles in the Moduli Space Bounds …

Entry e88b6be29f2d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-27 16:22:21 UTC

Minimal Rank of Algebraic Cycles in the Moduli Space Bounds Resolution Proof Depth

Entry ID: `e88b6be29f2d` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-27 16:22:21 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Moduli Spaces of Curves) **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

Statement:

> {'text': 'For every n-vertex planar graph G, let A(G) be the moduli space of complex algebraic curves with genus g such that G is embedded in the plane as a dual graph. Then, the minimal rank of an algebraic cycle representing a resolution proof for any 3-CNF instance F is upper-bounded by the rank of a generic algebraic cycle in A(G). Equivalently: For all 3-CNF instances F, max_{P∈R(F)} rank(C(P)) ≤ dim(A(G)), where R(F) is the set of resolution proofs for F.', 'quantitative': 'E[rank(C(P))] = O(dim(A(G)))', 'counterexample': 'A counterexample refutes the conjecture if there exists a 3-CNF instance F such that all generic algebraic cycles in A(G) have rank less than max_{P∈R(F)} rank(C(P)) for some resolution proof P of F.'}

Rationale (proposer's reasoning):

> {'text': 'The moduli space of curves provides a rich geometric structure that may encode the complexity of computations, such as the depth of resolution proofs. Exploring this connection could reveal new insights into the nature of computational hardness.', 'structural': 'By studying the geometric properties of moduli spaces, we may uncover invariants that correlate with computational complexity measures, potentially leading to new proof techniques for circuit lower bounds.'}

Taxonomy category: `MODULI_SPACE_RANK` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `aa48e28201366551`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all randomly generated 3-CNF instances with n ≤ 40 variables, the average rank of algebraic cycles in resolution proofs is within a polynomial

factor of the dimension of the moduli space A(G), and no instance has an algebraic cycle rank exceeding dim(A(G)). The conjecture is falsified if any instance exhibits an algebraic cycle rank greater than dim(A(G)) or if the average rank does not adhere to the polynomial bound.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "algebraic geometry moduli spaces curves" AND "resolution proof complexity" - "minimal rank algebraic cycles moduli space bounds" AND 3-CNF instance - "upper bound rank cycle resolution proof" IN PROOF

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=0, elapsed=0.0s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(-n, -1), random.randint(1, n)]
            if random.choice([True, False]):
                clause = [-x for x in clause]
            clauses.append(clause)
        return clauses

    def dual_graph(clauses):
        graph = {}
        for clause in clauses:
            for literal in clause:
                abs_literal = abs(literal)
                if abs_literal not in graph:
                    graph[abs_literal] = set()
                for other_clause in clauses:
                    if literal in other_clause and literal != -other_clause[0]:
```

```
                    graph[abs_literal].add(abs(other_clause[0]))
    return graph

def gaussian_elimination(matrix):
    n = len(matrix)
    rank = 0
    for i in range(n):
        if matrix[i][i] == 0:
            found_pivot = False
            for j in range(i + 1, n):
                if matrix[j][i] != 0:
                    matrix[i], matrix[j] = matrix[j], matrix[i]
                    found_pivot = True
                    break
            if not found_pivot:
                continue
        pivot = matrix[i][i]
        for j in range(i, n):
            matrix[i][j] /= pivot
        for k in range(n):
            if k != i and matrix[k][i] != 0:
                factor = matrix[k][i]
                for j in range(i, n):
                    matrix[k][j] -= factor * matrix[i][j]
        rank += 1
    return rank

def moduli_space_dimension(graph):
    # Simplified heuristic for the dimension of the moduli space
    # This is a placeholder and should be replaced with an actual computation
    return len(graph)

n = random.randint(5, 40)
clauses = generate_3cnf(n)
graph = dual_graph(clauses)
dim_A_G = moduli_space_dimension(graph)

rank_sum = 0
instances_tested = 10

for _ in range(instances_tested):
    resolution_proof = []
    for clause in clauses:
        literal = random.choice(clause)
        resolution_proof.append(literal)

    algebraic_cycle = [resolution_proof]
    rank_C_P = gaussian_elimination(algebraic_cycle)

    if rank_C_P > dim_A_G:
        return {
            "metric_name": "rank(C(P))",
            "metric_value": rank_C_P,
            "instances_tested": instances_tested,
            "conjecture_holds": False,
            "counterexample": f"Rank of algebraic cycle exceeds dimension of moduli space: {ra
```

```
            }

        rank_sum += rank_C_P

    mean_rank = rank_sum / instances_tested
    return {
        "metric_name": "E[rank(C(P))]",
        "metric_value": mean_rank,
        "instances_tested": instances_tested,
        "conjecture_holds": mean_rank <= dim_A_G * 2,  # Polynomial factor of 2 for simplicity
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] if sys.argv[1:] else [random.randint(1000, 9999) for _

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value:.2f} std={std_metric_value:.2f} support_
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Rank exceeds dimension\" first_failing_seed={fi
    else:
        print("RESULT: INCONCLUSIVE insufficient evidence")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
: 1.0, 'instances_tested': 10, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'E[rank(C(P))]', 'metric_value': 1.0, 'instances_tested': 10, 'conjecture_hol
TRIAL: {'metric_name': 'E[rank(C(P))]', 'metric_value': 1.0, 'instances_tested': 10, 'conjecture_hol
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```

```
RESULT: SUPPORTED mean=1.00 std=0.00 support_fraction=1.00
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> The empirical test only includes 10 instances, which is too small to be conclusive. The metric 'E[rank(C(P))] could scale trivially with n, and the conjecture might not hold for larger graphs.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> Safety rail: critic_challenge | original judge: The test results indicate that for all tested instances, the average rank of algebraic cycles in resolution proofs is within a polynomial factor of th | next: Further investigation with larger graphs and more instances to confirm the conjecture's validity across a wider range of cases.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14174
2	preregistration	ollama_remote	glm4:latest	0	0	6279
3	novelty	ollama_remote	glm4:latest	0	0	4854
4	novelty	ollama_remote	glm4:latest	0	0	5661
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	28527
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11429
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9701
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12273
9	critic	ollama_remote	glm4:latest	0	0	8990
10	judge	ollama_remote	glm4:latest	0	0	5715

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 107602 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/e88b6be29f2d.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e88b6be29f2d.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/e88b6be29f2d.tar.gz` (if generated)